

LA HW

Included exercises: 28, 30, 32, 55-59, 79-82, 91-92

Exercises 28-30

For Exercises 28–30, use the set

$$S = \{9 + 4x + 5x^2 - 3x^3, -3 - 5x - 2x^2 + x^3\}$$

Original Exercise 28

Prove that the polynomial $-6 + 12x - 2x^2 + 2x^3$ is a linear combination of the polynomials in S . Find the coef-

Original Exercise 30

cients of the linear combination.

Prove that the polynomial $8 + 7x - 2x^2 + 3x^3$ is *not* a

Original Exercise 32

Prove that

$$\text{Span} \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \right\} = \mathcal{M}_{2 \times 2}.$$

Original Exercise 55

Verify axiom 2 for $\mathcal{F}(S)$.

Original Exercise 56

Verify axiom 4 for $\mathcal{F}(S)$.

Original Exercise 57

Verify axiom 5 for $\mathcal{F}(S)$.

Original Exercise 58

Verify axiom 6 for $\mathcal{F}(S)$.

Original Exercise 59

Verify axiom 8 for $\mathcal{F}(S)$.

Original Exercise 79

multiplication by scalars are defined as in $\mathcal{F}(\mathcal{R})$.
Prove Theorem 7.2(b).

Original Exercise 80

Prove Theorem 7.2(d).

Original Exercise 81

Prove Theorem 7.2(f).

Original Exercise 82

Prove Theorem 7.2(h).

Original Exercise 91

is a subspace of W .

A function f in $\mathcal{F}(\mathcal{R})$ is called an **even function** if $f(t) = f(-t)$ for all t in $\mathcal{R}$ and is called an **odd function** if $f(-t) = -f(t)$ for all t in $\mathcal{R}$.

- (a) Show that the subset of all even functions is a subspace of $\mathcal{F}(\mathcal{R})$.
- (b) Show that the subset of all odd functions is a subspace of $\mathcal{F}(\mathcal{R})$.

Original Exercise 92

Let V be the set of all continuous real-valued functions defined on the closed interval $[0, 1]$.

- (a) Show that V is a subspace of $\mathcal{F}([0, 1])$.
- (b) Let W be the subset of V defined by

$$W = \left\{ f \in V : \int_0^1 f(t) dt = 0 \right\}.$$